

# Adherence to WCRF/AICR cancer prevention recommendations and risk of post-menopausal breast cancer

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**Background** In 2007 the World Cancer Research Fund (WCRF) and American Institute for Cancer Research (AICR) released eight recommendations related to body fatness, physical activity and diet aimed at preventing the most common cancers worldwide. However, limited information exists on the association between meeting these recommendations and risks of specific cancers, including breast cancer. **Methods** We operationalized six recommendations (related to body fatness, physical activity, foods that promote weight gain, plant foods, red and processed meats, and alcohol) and examined their association with invasive breast cancer incidence over 6.7 years of follow-up in the VITamins And Lifestyle (VITAL) study cohort. Participants included 30,797 post-menopausal women ages 50–76 years at baseline in 2000–2002 with no history of breast cancer. Breast cancers (n=899) were tracked through the Western Washington Surveillance, Epidemiology and End Results (SEER) database. **Results** Breast cancer risk was reduced by 60% in women who met at least five recommendations compared to those who met none (HR: 0.40; 95% CI: 0.25–0.65;  $P_{trend} < 0.001$ ). Further analyses that sequentially removed individual recommendations least associated with reduced risk suggested that this reduction is due to meeting recommendations related to body fatness, plant foods and alcohol (HR for meeting vs. not meeting these three recommendations: 0.38; 95% CI: 0.25–0.58;  $P_{trend} < 0.001$ ). **Conclusions** Meeting the WCRF/AICR cancer prevention recommendations, specifically those related to alcohol, body fatness and plant foods, is associated with reduced post-menopausal breast cancer incidence. **Impact** Increased adherence to the WCRF/AICR cancer prevention recommendations could substantially reduce post-menopausal breast cancer risk in US women.